



17 September 2026

DUNE Interface Control Document: Slow Controls, Photon Detector System and Detector Protection System

Summary

This document defines control, configuration and monitoring (CCM) interfaces for DAPHNE, the PDS Calibration Box and the PDS Power-over-Fiber (PoF) laser boxes between PDS, DAQ, Slow Controls and DPS. Each DAPHNE uses one shared 1 Gb Ethernet connection. DAQ CCM requests run and timing-endpoint configuration through the component integrated into the SC-provided external OPC UA server. SC can use that path to recover timing and publish board readiness. Hermes remains a separate service on the same link.

Configuration uses request/response; monitoring is published. DAQ opmon receives firmware counters, timing state and applied readout configuration from `daphne-server` through OPC UA. SC monitors voltages, temperatures, fans, service status and versions. The inventory comprises 42 FD-VD boards and 150 FD-HD boards. Calibration and PoF controllers expose equipment control and published monitoring through their own interfaces to OPC UA. DAQ data readout and optical timing transport are outside this scope.

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1 Responsibilities

Group	Role
PDS	Provides detector and calibration hardware, firmware, native services and specifications, including the PoF laser boxes.
SC	Provides control and monitoring infrastructure and arbitrates access to detector controls and monitoring data.
DPS	Provides protection infrastructure and detector protection.
DAQ	Uses the detector: requests run and calibration configuration and consumes monitoring and readiness through the supplied interfaces.

2 Physical interface and service placement

Each DAPHNE provides one dedicated 1 GbE SFP connection for CCM, as specified in the DAQ–PDS ICD [R1, R2]. Its board services share this link:

DAPHNE Ethernet interface $\longleftrightarrow$ switch $\longleftrightarrow$ physical server.

PDS supplies the fibres to its mini-rack patch panels; continuation and active switching are coordinated with DAQ and Technical Coordination [R1].

The PDS Calibration Box (Light Calibration Module, LCM) has its own dedicated 1 GbE SFP CCM connection [R2]. PDS PoF laser boxes connect through their laser controllers; their Ethernet medium, rate and connector remain to be specified. Both use the switched control network to reach the external physical server.

2.1 DAPHNE inventory

Detector	PDS channels	Channels/board	DAPHNE boards
FD-VD	1,344	32	42
FD-HD	6,000	40	150

These are the detector design allocations [R2], with one CCM connection per board. FD-VD includes membrane and cathode modules; both use this CCM interface. PoF supplies the cathode modules [R2]. PDS shall specify calibration-box and PoF-box quantities and their detector mapping separately from the board count.

2.2 Service locations

Location	Services
DAPHNE board	daphne-server and ZeroMQ endpoints; local timing-endpoint access; Hermes IPBus/UDP configuration service.
PDS Calibration Box	LCM controller and calibration control/monitoring service.
PDS PoF laser boxes	Laser controllers and control/monitoring endpoints; controller placement to be specified.
External physical server	SC-provided OPC UA server containing DAQ configuration, DAPHNE Protobuf/ZeroMQ client/subscriber, and adapters for calibration and PoF controllers.
SC server infrastructure	Ignition, acting as an OPC UA client of the external server.
DAQ server infrastructure	DAQ CCM client/opmon and the separate Hermes client.

Services may share a physical host. The installation record shall name their hosts, device/switch/server ports, media and patch routes. OPC UA terminates on the external server. DAPHNE services share each board’s available GbE bandwidth; calibration and PoF controllers have separate network endpoints.

3 Control, configuration and monitoring protocols

3.1 Service architecture

SC provides the OPC UA server and control arbitration. The server hosts DAQ configuration and forwards publications from **daphne-server**: equipment monitoring to Ignition; firmware counters, timing state and applied readout configuration to DAQ opmon. Hermes receives configuration over UDP/IPBus. Figure 1 shows the DAPHNE path; box interfaces are defined in Section 3.5.

3.2 Configuration exchanges

DAQ CCM requests run-variable and timing-endpoint configuration through OPC UA. The server exchanges Protobuf messages over ZeroMQ and TCP using a DEALER client and the board’s ROUTER endpoint [R3, R4]. The reference port is TCP 40001 [R5]. Correlated responses shall distinguish acceptance, completion and failure.

Hermes retains its independent IPBus/UDP configuration endpoint, reference UDP 50001 [R6], outside the OPC UA server. Both services share the board connection and gateway lifecycle.

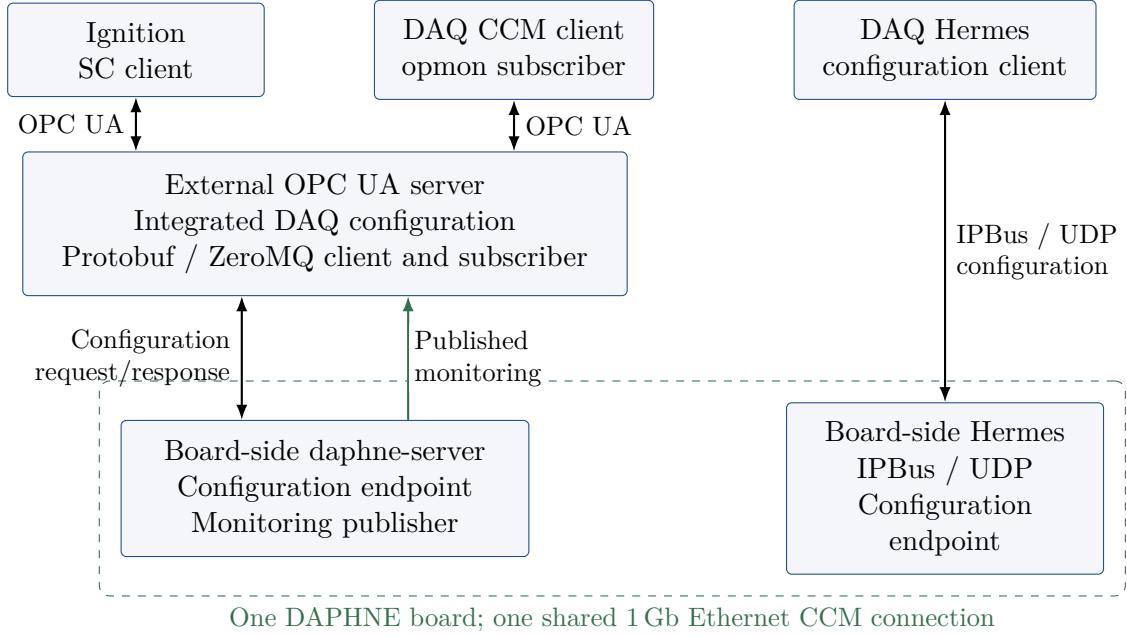


Figure 1: DAPHNE CCM configuration and published monitoring.

3.3 Published monitoring

Monitoring shall be published periodically or on change, without a client request for each update. Only configuration and control operations, including reset/recovery commands, use application request/response. Local hardware sampling is performed by the producing service.

The proposed DAPHNE publication transport is Protobuf over ZeroMQ PUB/SUB over TCP, using endpoints separate from configuration. The OPC UA server subscribes to `daphne-server` publications and distributes them through OPC UA subscriptions. Opmon monitoring follows the path `daphne-server` → OPC UA server → DAQ opmon.

Consumer	Monitoring responsibility
DAQ opmon	Firmware counters, timing state and applied data-readout configuration, published by <code>daphne-server</code> through OPC UA.
SC / Ignition	Voltages, temperatures, fans, service status and software/firmware versions.
SC and DAQ	Timing-endpoint condition, reset/recovery progress and board readiness.

Opmon shall not acquire voltage monitoring. SC receives equipment monitoring through the OPC UA server without an opmon dependency. Publications shall carry source identity, time, sequence and validity. Subscribers shall detect stale or missing updates and obtain a current snapshot after reconnecting.

3.4 Timing-endpoint control and recovery over GbE

Timing-endpoint configuration follows the DAQ configuration path:

DAQ $\longleftrightarrow$ OPC UA server $\longleftrightarrow$ DAPHNE service $\longleftrightarrow$ local timing endpoint.

The same interface shall expose reset and recovery to authorized SC clients. Timing monitoring shall be published over the board's GbE connection and made available to both SC and DAQ.

A reset request shall identify its target and receive an execution response. The board shall publish reset assertion, release and the resulting endpoint state so that both clients can acknowledge the same recovery operation. Recovery access shall remain available while timing is unavailable. Concurrent DAQ and SC commands shall be serialized by SC's control infrastructure.

SC shall determine and publish *board ready for data taking* from the agreed readiness conditions, including completed reset, restored clocks, endpoint readiness and valid synchronization [R4]. Reset release alone is insufficient. Missing or stale required publications shall withhold readiness. DAQ consumes the published readiness before resuming data taking.

3.5 PDS Calibration Box and PoF laser boxes

Each box controller shall expose configuration/control request/response and published monitoring through its native service and an adapter in the external OPC UA server. Ignition subscribes to equipment monitoring; DAQ subscribes to the readiness and operating status needed for data taking.

Calibration Box (LCM). PDS supplies the calibration hardware, firmware and service. DAQ uses that service to request calibration settings and sequence execution. SC provides command arbitration, equipment power/service control and monitoring of health, alarms and calibration status. The OPC UA interface shall expose protective inhibit actions.

PoF laser boxes. SC controls optical-power operation and receives controller health, operating state, permits and inhibit status. The interface shall expose DPS protection status and give protective inhibits priority over operating commands. PDS and DPS shall define the controller/DPS connection and its acknowledgement behavior.

PDS shall publish each controller's native protocol, transport, ports, command acknowledgements, monitoring schema and compatible versions. These details remain to be specified for both box types; the DAPHNE Protobuf/ZeroMQ contract applies only to DAPHNE.

3.6 Interface release and verification

Each release shall publish compatible versions and interface specifications online: schemas, OPC UA model, Hermes and box-controller interfaces, and endpoint/port allocation. Contracts shall define authentication, timeouts, queue limits, update cadence and reconnect behavior. DAPHNE publication schemas and ports remain to be assigned.

Verification shall cover concurrent configuration and publication on the shared link, the HD and VD installation scales, subscriber loss/reconnection, and SC timing recovery with a board-generated completion acknowledgement. Box verification shall cover command ownership, published status, reconnect behavior and inhibit acknowledgement. The DAQ-PDS and DAQ-SC/DPS interface documents shall use this same CCM contract.

References

- R1** DAQ-PDS ICD, 2621471: CCM physical connection.
- R2** DAQ-PDS proposed v8, 6c0b245: format, HD/VD inventory, configuration architecture, LCM CCM interface and ownership, cathode PoF supply.
- R3** daphnmodules, edf75a0: configuration schema and DEALER client.
- R4** daphne-os/daphne-server, e31bdfb: ROUTER configuration endpoint and endpoint recovery prerequisites.
- R5** daphne-os/petalinux, e31bdfb: board service endpoints and lifecycle.
- R6** hermesmodules, 887d5fc: independent IPBus/UDP configuration service.